

Fixed Points VI: Formal Synthesis and Physical Interpretations

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Version 7

Abstract

We synthesize the canonical FP–I–V results without promoting their control parameter, projections, or diagnostics into an unproved fundamental field theory. The rigorous spine consists of conditional projected fixed-point existence, strict-Lyapunov promotion to equilibrium, joint-mode damping and disturbance floors, perturbative persistence of a curved spectral cluster, curvature leakage, intrinsic first-order centroid modulation, frozen linear Ornstein–Uhlenbeck covariance, canonical-correlation bounds, and declared admissibility exits. We distinguish three logical levels throughout: inherited theorem, conditional model completion, and physical interpretation. All inherited results are restated as self-contained contracts and cited to their canonical paper; FP VI does not reproduce their theorem/proof blocks. Its own formal results are limited to the bilocal-causality obstruction and the local-mediator completion. A Lorentzian gauge/gravity action, quantum covariance, particle identities, merger, measurement, and cosmology belong to the latter two levels unless additional source and equivalence theorems are supplied. The Gaussian Lyapunov sign and nonnormal resolvent bounds are stated in their domain-valid forms, and we show why an instantaneous equal-time bilocal kernel is not microcausal merely because it leaves the principal symbol unchanged. A local mediator is the consistent route to causal overlap dynamics. The synthesis uses only the preceding Fixed Points papers and standard mathematical literature. Physical interpretations are conditional applications, not additional proof inputs.

Revision note for version 7

Supersedes. Version 6, retained as the released edition.

Reason. Even an explicitly separate downstream progress ledger placed later research inside a foundational synthesis and introduced reverse citations.

Resolution. The ledger and application-specific carrier claims are removed. The source chain is now FP I–V only, alongside standard mathematics; the cited FP editions are aligned with the present foundational revisions.

Retained result. The analytic synthesis, its stated assumptions, and the two bilocal-causality propositions retain their roles.

Remaining boundary. Concrete physical dynamics and their relation to the control framework must be established by applications of this foundation.

Revision note for version 6

Supersedes. *Fixed Points VI: Formal Synthesis and Physical Interpretations*, version 5.

Reason. Version 5 correctly separated the FP spine from later q79 work, but the synthesis still needed a clearer reading route and firmer protection against treating restated contracts as duplicate theorem sources.

Resolution. Version 6 organizes the paper into rigorous, downstream, and interpretive reading lanes; explains why contracts are repeated while proofs are not; and preserves FP I–V as the canonical owners of their results. It also separates release description, revision history, and computational provenance.

Retained result. The inherited FP I–V spine and the two FP VI bilocal-causality results retain exactly their version 5 scope.

Remaining boundary. The selected action, physical q79 continuum operator, general Born source, fixed-coupling QFT, global state, matching, and cosmology remain downstream obligations.

Revision note for version 5

Supersedes. *Fixed Points VI: Formal Synthesis and Physical Interpretations*, version 4.

Reason. Version 4 correctly narrowed the FP series, but it predates the exact q79 lane-factor, projective-module, connection-compiler and nonlinear strain-quotient results, as well as later scoped QM and QFT closures. Without a separate status layer, “not derived by FP” could be misread as “not achieved anywhere downstream.”

Resolution. Version 5 preserves the FP theorem boundary, corrects the q79 preprojection types, and adds an independently sourced downstream ledger with exact closure counts and nonpromotion clauses. It also restores strict theorem ownership: FP I–V retain their respective fixed-point, damping, curvature, and covariance theorems, while FP VI owns only its two new bilocal-causality results.

Retained result. The complete FP–I–V fixed-point, damping, curvature, covariance, and admissibility spine survives at its declared theorem tier.

Remaining boundary. Selected HYM endpoints and action, the physical q79 Hessian and reduction, universal apparatus and actualization, a fixed-coupling continuum QFT, selected global state, RG and matching, and cosmology remain downstream programs.

How to use this synthesis

This paper is a map of the Fixed Points chain, not a substitute for its five source papers. It restates the input–output contracts needed to understand the analytic sequence and cites each earlier canonical owner. Its only new formal results concern the causality failure of instantaneous bilocal overlap and the local-mediator completion.

The central picture in plain language. The series builds a sequence of conditional gates. Geometry and spectral data define a coherent projector. Compactness or condensation gives a projected return. A strict Lyapunov law promotes that return to equilibrium. Spectral margins control disturbances; curvature can rotate the projector and source leakage; covariance describes a frozen linear fluctuation cloud; and declared margins detect exit from a chosen admissible region. Each gate has its own hypotheses, and passing one does not silently pass the next.

Reading route. Sections 1–7 assemble the analytic results and their input contracts. Sections 8–9 discuss an explicitly conditional local action and the causality of overlap interactions. Section 10 separates possible physical interpretations from the inherited mathematics, and Section 11 summarizes the full dependency chain. No application-specific result is required to read or establish that chain.

Why contracts are repeated but proofs are not. A standalone synthesis must state enough hypotheses and conclusions to be understood without opening five documents at once. It therefore repeats short contracts, not theorem ownership. Proofs and canonical formal statements remain in FP I–V. This prevents the same theorem from acquiring multiple apparent sources while preserving standalone readability.

Scope boundary. The control flow, classical covariance, and admissibility diagnostics do not by themselves define Lorentzian dynamics, quantum commutation relations, particle identities, or cosmology. Applications may use these results after checking their hypotheses; their additional conclusions are not premises of the Fixed Points theorems.

Contents

Revision note for version 6	1
Revision note for version 5	2
How to use this synthesis	2
1 Logical levels and common geometry	4
2 Fixed-point and equilibrium synthesis	4
3 Joint-mode stability and disturbance floors	5
4 Curvature, centroid motion, and interaction	5
5 Frozen Gaussian multi-structure theory	6
6 Classical covariance versus quantum covariance	6
7 Admissibility and exit	7
8 Status of a Lorentzian master action	7
9 Bilocal overlap and causality	8
10 Physical interpretation ledger	8
11 Synthesis ledger and remaining obligations	9
12 Conclusion	9

1 Logical levels and common geometry

Every statement in this synthesis has one of the following statuses.

Inherited theorem. A consequence of FP I–V under hypotheses restated or cited here.

Conditional completion. A mathematically specified extra model whose conclusions hold if that model and its hypotheses are adopted.

Physical interpretation. A proposed reading that is not established by the fixed-point theorems alone.

Let $\mathcal{H}$ be the declared control Hilbert space. The joint internal operator is constructed from three declared vertical operators using the strong-commutation and domain hypotheses of FP II. Circle–lens–nil labels refer here to obstruction, rank, or operator data; they do not assert a literal product or nesting of three manifolds. A common $U(1)$ phase may be represented by a line connection without adding a product dimension, as in FP II. Finite rank flags do not specify differential operators or a Galerkin basis. Any such realization must be supplied before invoking the analytic estimates; no selected compactification or physical carrier is assumed here.

The coherent projector P is the joint spectral projector. A product of three bundle projectors is permitted only when those projectors strongly commute; otherwise the product need not be an orthogonal projector. Write $Q = I - P$.

On a curved bundle, the normalized Laplace-type operator is

$$L_x = \nabla_x^* \nabla_x + \mathcal{R}_x.$$

When the selected low cluster remains separated, FP IV defines the curved Riesz projector $P_{\mathcal{R}}(x)$. Retaining the old projector P instead requires the off-diagonal leakage block QRP .

The parameter of the FP gradient flow is a stabilization or control parameter. Identifying it with Lorentzian proper time, coordinate time, or a quantum evolution parameter is a conditional completion, not an inherited theorem.

2 Fixed-point and equilibrium synthesis

Let Φ_τ be a well-defined time- τ map for the declared control flow and set $T = P\Phi_\tau$ on a closed bounded convex set $K \subset \mathcal{H}$.

Imported fixed-point chain. Fixed Points I and II are the canonical sources for this chain [1, 2]. For standalone use, its complete contract is as follows. Assume one of the following FP I alternatives:

1. $T(K) \subset K$ and $T : K \rightarrow K$ is continuous and compact; or
2. $T(K) \subset K$ and T is continuous and condensing for a declared measure of noncompactness.

Then T has a fixed point $u_* \in K$. This proves $P\Phi_\tau(u_*) = u_*$. If, in addition, the trajectory through u_* remains in the projected invariant set and obeys a strict Lyapunov identity

$$\mathcal{C}(u(\tau)) - \mathcal{C}(u(0)) = - \int_0^\tau \mathcal{D}(u(s)) ds, \quad \mathcal{D} \geq 0,$$

with $\mathcal{D}(u) = 0$ exactly at equilibria, then u_* is an equilibrium of the declared flow. The first conclusion is an application of Schauder’s or Darbo–Sadovskii’s theorem. At a projected

recurrent point, the Lyapunov values at the two endpoints agree. The displayed identity forces $\mathcal{D} = 0$ along the intervening orbit, and strictness gives equilibrium. Without the second step, a projected time-step fixed point is not automatically a steady solution.

Imported contraction-uniqueness gate. Fixed Points I owns the generic Banach gate. In the steady-equation form used here, suppose a steady equation can be written on a specified Banach space as $u = -A^{-1}N(u)$, where A^{-1} exists on that space and

$$\|A^{-1}\| \text{Lip}(N) < 1$$

on an invariant complete subset. Then the steady solution there is unique. No uniqueness conclusion follows from a vertical spectral gap if A has an unremoved kernel or if the displayed inverse estimate is not proved.

3 Joint-mode stability and disturbance floors

Let α be one joint modal label, including multiplicity. After the FP III remainder estimate [3], suppose a noncoherent amplitude satisfies

$$\frac{d}{dt}|a_\alpha| \leq -\gamma_\alpha|a_\alpha| + |f_\alpha(t)|, \quad \gamma_\alpha > 0.$$

Then the deterministic input-to-state floor is $\|f_\alpha\|_\infty/\gamma_\alpha$. For the exact scalar Itô equation

$$da_\alpha = -\gamma_\alpha a_\alpha dt + \sqrt{q_\alpha} dW_\alpha,$$

the stationary variance is $q_\alpha/(2\gamma_\alpha)$. Deterministic force amplitude and stochastic power have different units and cannot be represented by one shared disturbance threshold.

For infinitely many modes, bundlewise stochastic control requires the appropriate weighted covariance trace; deterministic control requires its own weighted series. These are sufficient nonlinear stability criteria. Gaussian invariant laws are asserted only for exact linear OU systems.

4 Curvature, centroid motion, and interaction

Use the curved-operator framework of FP IV [4]. Let the uncurved noncoherent spectrum start at $\lambda_* > 0$. If $\|\mathcal{R}\| < \lambda_*/2$, the selected low spectral cluster remains separated and has a curved Riesz projector of the same rank. If the uncurved projector is retained and the one-sided Q -sector damping margin is $\gamma_Q > 0$, FP IV gives the schematic sharp form

$$\frac{d}{dt}\|q\| \leq -\gamma_Q\|q\| + \|\mathcal{Q}\mathcal{R}P\| \|p\| + \|\mathcal{Q}F\|,$$

and therefore a curvature-induced leakage floor.

For a localized profile inside a strongly convex normal ball, the centroid is the Karcher mean. A first-order gradient parent flow yields a first-order modulation equation

$$G_{ij}(X)\dot{X}^j = -\partial_i V_{\text{eff}}(X) + \varepsilon_i.$$

A Newton equation requires a separately declared inertial parent theory.

For two profiles, absolute cross-term control proves only

$$|E_{\text{int}}| \leq C\mathcal{O}.$$

Attraction requires $E_{\text{int}} \leq -c\mathcal{O}$; repulsion requires the opposite sign. Even a proved attractive sign does not establish merger, collapse, or a transition rate without a dynamical connection theorem.

5 Frozen Gaussian multi-structure theory

Consider the real linear SDE

$$d\zeta = A\zeta dt + B dW_t, \quad Q := BB^\top. \quad (5.1)$$

Imported semigroup-covariance result. Fixed Points V is the canonical source for the nonnormal covariance theorem [5]. For the real linear SDE (5.1), assume

$$\|e^{tA}\| \leq Me^{-\omega t}, \quad t \geq 0, \quad M \geq 1, \quad \omega > 0.$$

Then (5.1) has stationary covariance

$$\Sigma = \int_0^\infty e^{tA} Q e^{tA^\top} dt,$$

which solves

$$A\Sigma + \Sigma A^\top + Q = 0.$$

Moreover,

$$\|\Sigma\| \leq \frac{M^2\|Q\|}{2\omega}, \quad \|A^{-1}\| \leq \frac{M}{\omega}.$$

The semigroup estimate makes the covariance and resolvent integrals converge. Differentiation of the covariance integrand gives the Lyapunov equation, while $A^{-1} = -\int_0^\infty e^{tA} dt$. Taking norms proves the bounds.

For a normal A , one may often take $M = 1$ with ω given by the spectral abscissa. For a nonnormal matrix, the spectral abscissa alone does not imply $\|A^{-1}\| \leq 1/\omega$ or $\|\Sigma\| \leq \|Q\|/(2\omega)$; transient amplification must be controlled.

Under the block-diagonal damping assumptions of FP V, the cross covariance satisfies a Sylvester equation. Canonical correlation is normalized by the smallest positive eigenvalues of the marginal covariance blocks, not their largest operator norms.

6 Classical covariance versus quantum covariance

The word ‘‘covariance’’ occurs in both classical probability and quantum theory, but the objects satisfy different constraints. The FP V Lyapunov equation produces ordinary second moments of a classical stochastic process. A quantum covariance becomes physical only after a symplectic or canonical commutation structure and compatible completely positive dynamics have been specified.

The covariance produced by (5.1) is classical. To interpret it as a bosonic quantum covariance one must separately specify canonical commutation relations, a value of $\hbar$, and quantum dynamics whose noise and damping satisfy the complete-positivity constraints. Only then is

$$\Sigma + \frac{i\hbar}{2}\Omega \succeq 0$$

the physical uncertainty condition. Classical positivity of Σ does not imply this inequality.

For a declared bipartite Gaussian quantum state, nonpositivity after partial transpose certifies entanglement. PPT is exact for the two-mode 1×1 case [7]; in general multimode bipartitions PPT need not be sufficient for separability. These facts classify an already constructed quantum Gaussian state. They do not quantize the FP control flow or derive particle statistics.

7 Admissibility and exit

Let $m_j(x)$, $j \in \mathcal{J}$, be a finite family of continuous declared margins: curved cluster separation, one-sided damping, or explicit deterministic/stochastic performance tolerances. Define

$$\mathcal{D}_\varepsilon = \{x : m_j(x) \geq \varepsilon \ \forall j\}, \quad \mathfrak{B}(x) = \max_j [-m_j(x)].$$

Then $x \in \mathcal{D}_0$ exactly when $\mathfrak{B}(x) \leq 0$. Along a continuous path, a change from $\mathfrak{B} \leq 0$ to $\mathfrak{B} > 0$ has a first boundary time at which $\mathfrak{B} = 0$.

This is an exact exit diagnostic for the declared constraints. It does not prove that $\mathfrak{B}$ is a physical energy or force. It does not establish which state follows exit. A mountain-pass theorem needs an actual energy functional, and basin selection needs a post-exit evolution or transition kernel.

For affine observables of a frozen Gaussian system, FP V supplies finite-grid and Borell–TIS continuous-time exit estimates. A Lipschitz nonlinear function of a Gaussian process is not generally Gaussian. Equal-time cross-correlation can bound simultaneous observed exits, but cannot prove causal propagation or non-propagation.

8 Status of a Lorentzian master action

This section illustrates a conditional completion. Writing a familiar local action shows what additional fields and equations could supply physical evolution; it does not show that the FP control data select that action. The gap between a writable action and a derived action is precisely the missing source and matching problem.

A local Lorentzian gauge/gravity/matter action may be appended as a *conditional completion*, for example schematically

$$S_{\text{cand}} = \int \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \Lambda - \frac{1}{4} \sum_r \text{tr } F_r^2 - \sum_a |D\varphi_a|^2 - V(\varphi) + \sum_b \bar{\psi}_b (i\gamma^\mu D_\mu - M_b) \psi_b \right] d^4x.$$

The FP I–V theorems do not select the gauge group, representations, number of fields, couplings, Yukawa matrices, or potential. Nor do they prove that the gradient control flow is the Euler–Lagrange evolution of S_{cand} .

If a mass term depends on curvature, such as $M_a^2 = m_a^2 + \xi_a R$, its metric variation contributes nonminimal terms to the gravitational field equation; it cannot be inserted while retaining the minimally coupled Einstein equation unchanged. Likewise, a point-particle Lorentz-force law requires a controlled localized-solution limit and is not a consequence of gauge covariance alone.

Three internal structures do not by themselves prove three gauge factors, bosonic or fermionic statistics, bifundamental matter, the Standard Model particle assignment, anomaly cancellation, or equivalence with a quantum field theory. Those require independent representation, source, and matching theorems.

9 Bilocal overlap and causality

Consider an equal-parameter interaction of the form

$$\mathcal{N}[\phi](t, x) = \int_{\Sigma_t} K(x, y) |\phi(t, y)|^2 d\mu_t(y) \phi(t, x).$$

Proposition 9.1 (Instantaneous bilocal obstruction). *If $K(x, y)$ is nonzero for spatially separated x and y , the equation at (t, x) depends instantaneously on data at y . Leaving the differential principal symbol unchanged is therefore insufficient to prove the usual local domain-of-dependence property. An L^1 bound on K may help local well-posedness, but it does not establish microcausality.*

Proof. Choose two initial data sets that agree near x but differ near a separated point y where $K(x, y) \neq 0$. Their bilocal source values at (t, x) differ at the same parameter time. Thus the source is not determined by data in an arbitrarily small causal neighborhood of x . $\square$

Proposition 9.2 (Local-mediator completion). *Introduce a local mediator χ with a hyperbolic equation, for example*

$$(\square_g + m_\chi^2)\chi = g_\chi |\phi|^2,$$

and couple χ locally back to ϕ . Under the standard regularity, gauge, and constraint hypotheses for the resulting local hyperbolic system, its domain of dependence is governed by the local characteristics. Eliminating χ with a retarded Green operator produces a causal retarded memory kernel, not in general a symmetric equal-time bilocal action.

This local parent theory is the preferred route if overlap dynamics is meant to represent physical causal interaction.

10 Physical interpretation ledger

The ledger should be read as a type system for claims. “Supported” means the stated control model and hypotheses already imply the claim. “Conditional” means an additional mathematical model would make the claim well posed and testable. “Not derived” means the Fixed Points chain alone contains no implication to that conclusion; it is not a declaration that the idea is impossible or absent from every other MTT paper.

Supported within the declared control model. The declared model supports projection into coherent and noncoherent sectors under operator-domain hypotheses. It also supports conditional fixed points, strict-Lyapunov equilibrium promotion, damping estimates and disturbance floors, perturbative curved-cluster persistence, curvature leakage, intrinsic first-order centroid modulation, frozen linear covariance, and admissibility exit diagnostics.

Conditional after adding a model. Gauge forces, Lorentzian motion, quantum uncertainty, Gaussian entanglement, decoherence, thermodynamic entropy production, and causal overlap can be studied after a compatible local Lorentzian or open-quantum model is specified. Their parameters and consistency conditions are additional data.

Not derived by this series. The present theorems do not derive boson/fermion identity, particle masses, Standard Model representations, merger or collapse rules, the Born rule, measurement outcomes, emergent time, inflation, horizons, or a cosmological arrow of time. This is a statement about the implications of this series, not a claim about the status of every physical theory or application.

11 Synthesis ledger and remaining obligations

Imported FP synthesis.

Assume the operator, compactness or condensing, strict-Lyapunov, one-sided damping, curvature-smallness, covariance, and margin hypotheses stated in FP I–V and summarized above. Then the declared control model has:

1. a projected time-step fixed point, promoted to an equilibrium only by the strict Lyapunov condition;
2. quantified noncoherent damping and deterministic/stochastic floors;
3. a persistent curved low cluster and a quantified old-projector leakage term;
4. intrinsic first-order centroid modulation under the modulation hypotheses;
5. exact stationary covariance for the frozen linear OU sector and correctly normalized correlation bounds; and
6. exact detection, plus conditional Gaussian probability bounds, for exit from a declared admissible domain.

No Lorentzian, quantum, particle-physics, merger, measurement, or cosmological claim follows without its additional completion theorem.

For an application, the constructive obligations are to specify an energy and its operator domain, establish the invariant set and relevant analytic bounds, and prove the relation between the control flow and the proposed physical evolution. Quantum or relativistic interpretations need their own state, causality and observable structures. The present paper neither supplies those structures nor needs them to establish the analytic synthesis.

12 Conclusion

The corrected Fixed Points series is a coherent conditional control and spectral framework. Its strongest results concern existence under explicit compactness hypotheses, stability margins, curved spectral persistence, leakage, modulation, Gaussian covariance, and admissibility diagnostics. FP VI does not weaken those results by surrounding them with broader interpretations; it identifies exactly what they prove and exactly which bridges remain before they can support a fundamental spacetime, quantum, or particle theory.

The synthesis also clarifies how to use the series. FP I owns the analytic fixed-point foundation; FP II owns the ten-dimensional joint-operator specialization; FP III owns disturbance floors and conditional homogenization; FP IV owns curvature, finite reduction, centroid, and transition boundaries; FP V owns multi-structure covariance and admissibility-exit probabilities. FP VI connects those contracts and adds only the bilocal-causality analysis. The next advances must therefore instantiate the missing source data or physical completion, not restate the existing fixed-point chain under a new heading.

References

- [1] P. Nero, *Fixed Points I: Fixed Points over Multi-Bundle Manifolds*, version 7, 2026.
- [2] P. Nero, *Fixed Points II: Projected Fixed Points and Equilibria in a 10D Modal Model*, version 6, 2026.
- [3] P. Nero, *Fixed Points III: Disturbance-Damping Balance and Stability*, version 8, 2026.
- [4] P. Nero, *Fixed Points IV: Curvature, Centroid Motion, and Structural Transitions on Bundle Manifolds*, version 7, 2026.
- [5] P. Nero, *Fixed Points V: Curvature Coupling, Multi-Structure Dynamics, and Admissibility Barriers*, version 9, 2026.
- [6] G. Da Prato and J. Zabczyk, *Stochastic Equations in Infinite Dimensions*, Cambridge University Press, 1992.
- [7] R. Simon, Peres-Horodecki separability criterion for continuous variable systems, *Physical Review Letters* **84** (2000), 2726–2729.

Dependency and reproducibility statement

The preceding FP papers are the sole MTT proof sources used here. Standard mathematics and the two local causality arguments complete the synthesis. No later research ledger, numerical packet or physical endpoint is needed.